2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)
31 July – 2 August 2025, Rangpur, Bangladesh

Explainable Machine-Learning Framework for Predicting Delivery Delays and Risk in Smart Supply Chains

Khandaker Fariha Islam
Dept. of CSE
East West University
Aftabnagar, Dhaka, Bangladesh
k.fariha.islam15@gmail.com

Md Ripon Al Mamun
Dept. of CSE
East West University
Aftabnagar, Dhaka, Bangladesh
riponalmamunrasel@gmail.com

Mohammad Rifat Ahmmad Rashid
Dept. of CSE
East West University
Aftabnagar, Dhaka, Bangladesh
rifat.rashid@ewubd.edu

Sadia Chowdhury
Dept. of CSE
East West University
Aftabnagar, Dhaka, Bangladesh
tumpamahbub@gmail.com

Raiyan Gani
Dept. of CSE
East West University
Aftabnagar, Dhaka, Bangladesh
raiyan.gani02@gmail.com

Raihan Ul Islam
Dept. of CSE
East West University
Aftabnagar, Dhaka, Bangladesh
raihan.islam@ewubd.edu

Meherun Nessa Jerin
Dept. of CSE
East West University
Aftabnagar, Dhaka, Bangladesh
meherunnessajerin@gmail.com

Jubaer Ahmed
Dept. of CSE
East West University
Aftabnagar, Dhaka, Bangladesh
jubaerahmed038@gmail.com

M. Saddam Hossain Khan
Dept. of CSE
East West University
Aftabnagar, Dhaka, Bangladesh
saddam.cse@ewubd.edu

***Abstract*—Modern supply chains face rising pressure to anticipate late shipments, mitigate delivery-risk and respond to volatile customer demand. Leveraging a public 182 020-record "SMART Supply-Chain" data set, we couple classical statistics with explainable machine learning (ML) to pinpoint the factors that most strongly drive delivery outcomes and sales behaviour. Paired–t tests, Pearson correlation and χ^2 analysis reveal: (i) a highly significant gap between scheduled and actual shipping days ($p < 10^{-4}$), (ii) a strong positive link between product price and customer sales ($r = 0.78$), and (iii) shipping-mode and delivery-status associations with late-delivery risk ($p < 10^{-4}$). Six supervised learners—Linear, Ridge and Lasso regression, Random Forest, Decision Tree and Gradient-Boosting—are then benchmarked. Ridge regression attains the best predictive performance (MSE $= 2.27\times10^{-6}$, MAE $= 5.7\times10^{-4}$, $R^2 = 0.9999$) while maintaining model parsimony. To ensure managerial transparency, we generate instance-level explanations with SHAP and LIME, showing that *Order-Item-Total*, *Sales-per-Customer* and *Order-Item-Discount* are the principal drivers of the ridge model's risk forecasts. The integrated framework supplies actionable insight as well as near-perfect accuracy, enabling logistics teams to prioritise high-risk consignments, fine-tune inventory policy and refine shipping-mode selection.**

***Keywords*—Supply Chain, Machine Learning, Delivery Risk, Predictive Analytics, Explainable AI.**

I. Introduction

The globalisation of commerce, shortening product lifecycles, and escalating customer expectations have made today's supply chains both data-intensive and highly disruption-prone. Late deliveries propagate across supplier tiers, inflating costs, eroding goodwill, and undermining overall resilience. Traditional rule-based or purely descriptive analytics struggle to uncover the multifaceted causes of delay—from adverse weather and congested ports to inventory imbalance and sub-optimal shipping modes. Consequently, industry leaders are embracing *machine learning* (ML) for predictive risk assessment and adaptive decision support.

Recent research confirms that learning algorithms can mine historical shipment records to forecast lead-time variability more accurately than classical forecasting tools. Two persistent gaps, however, limit practical impact. First, many pipelines optimise prediction without statistically validating the relationships between operational variables and observed delays. Second, the highest-performing models are often opaque "black boxes," offering limited managerial insight into *why* a consignment is deemed high-risk—reducing trust and slowing adoption.

This study closes these gaps by coupling rigorous hypothesis testing with an *explainable* ML framework. Using the public *DataCo SMART Supply Chain* data set (182 020 records, 53 attributes), we (1) quantify statistical dependencies among scheduled and actual shipping days, product/customer characteristics, and late-delivery risk; (2) benchmark six supervised learners—Linear, Ridge, and Lasso regression, Random

2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN) | 979-8-3315-9694-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/QPAIN66474.2025.11171898

979-8-3315-9694-1/25/$31.00 ©2025 IEEE

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on April 16,2026 at 04:01:33 UTC from IEEE Xplore. Restrictions apply.

Forest, Decision Tree, and Gradient-Boosting Regressor—for delay prediction; and (3) employ SHAP and LIME to deliver instance-level explanations that reveal the operational drivers of each forecast.

Objectives of this Work

- Develop and validate data-driven models that accurately predict delivery delays in a smart-supply-chain context.
- Statistically confirm the key factors—such as shipping mode, product attributes, and customer behaviour—that influence late-delivery risk.
- Provide transparent, instance-level explanations of model predictions using state-of-the-art explainable-AI techniques (SHAP and LIME) to support actionable decision-making.

By uniting robust statistical inference with high-fidelity, interpretable prediction, the proposed framework equips logistics managers to pre-empt late shipments, allocate mitigation resources effectively, and refine shipping-mode strategies—thus advancing the practical integration of AI in modern supply-chain management.

II. Related Work

A. AI–enabled Resilience and Visibility

Belhadi *et al.* [1] applied a multi-criteria decision-making (MCDM) framework to data from 479 manufacturing firms, showing that wavelet neural networks, fuzzy logic and big-data analytics jointly enhance supply-chain resilience (SCRes) by improving adaptability, response time and risk assessment. Complementing this, Rejeb *et al.* [2] surveyed 43 studies and demonstrated that augmented-reality (AR) applications in warehousing, manufacturing and logistics bolster real-time visualisation and operational efficiency; they recommend further work on AR–AI–IoT convergence.

B. Predictive Risk-Mitigation Models

With a focus on financial exposure, Lei *et al.* [3] proposed a four-stage pipeline—pre-processing, Chaotic Grasshopper Optimisation for feature selection, SVM classification and Slime-Mould parameter tuning—that outperformed traditional baselines in both accuracy and decision-making speed. Aljohani [4] argued for continuously updated ML dashboards that learn from contextual and historical data, reporting gains in forecast accuracy and reaction time.

C. Demand Forecasting and Sustainability

Cross-series transfer of time-series signals, smart grouping and downstream-inventory features enabled Zhu *et al.* [5] to cut mean-squared-error (MSE) and mean-absolute-error (MAE) in pharmaceutical demand forecasts. Khan *et al.* [6] surveyed 290 SMEs in Pakistan and China, finding that digital technologies drive circular-economy practices and boost economic as well as environmental KPIs.

D. Explainability, Blockchain and Decision Support

Nimmy *et al.* [7] highlighted the dominance of black-box predictors in operational-risk models and called for the systematic adoption of explainable-AI (XAI) techniques—measured via F_1, recall, accuracy and precision—to improve managerial trust. Blockchain-centred visibility frameworks for third-party logistics were outlined by Tiwari *et al.* [8], who linked cybersecurity and data-quality hurdles to supply-chain security, transparency and efficiency metrics.

E. AI in Operations and Performance

An experimental case study by Helo and Hao [9] illustrated AI's benefits for process optimisation, inventory control and demand forecasting in supply-chain and operations management. Maaz and Ahmad [10] surveyed 1,139 dairy-industry stakeholders, confirming that superior supply-chain performance—mediated by customer satisfaction—translates into improved organisational outcomes.

F. Visibility, Intellectual-Property and Digitalisation Risks

Hunaid *et al.* [11] identified delivery visibility and technology integration as pivotal for supply-chain visibility (SCV) in shipping companies, thereby supporting faster decision-making. Rauf *et al.* [12] underscored big-data analytics for early detection of intellectual-property (IP) threats, yet noted challenges such as data quality and model interpretability. Zhou *et al.* [13] combined case studies and surveys to show that AI, IoT and blockchain strengthen agility, risk mitigation and operational efficiency, though cybersecurity and talent shortages remain barriers. Building on trust dynamics, Zhou, Wang and Yang [14] used mathematical modelling for multi-criteria sorting, tracking decision accuracy and consensus efficiency but cautioned that static-trust assumptions may limit real-world applicability. Finally, Tao *et al.* [15] demonstrated—at a theoretical level—that effective risk management can buffer supply-chain performance against demand unpredictability, urging empirical validation with real-world data.

G. Research Gap

Collectively, prior studies confirm the value of AI, blockchain and XAI for supply-chain optimisation, yet delivery-delay prediction with *instance-level* explainability remains under-explored. Our work addresses this gap by: (i) modelling delivery status as an explicit target, (ii) validating statistical dependencies before learning, and (iii) providing SHAP- and LIME-based explanations to bridge the divide between high-accuracy prediction and actionable insight.

III. Methodology

The first step demanded a review of the entire dataset to establish how data was distributed between columns in addition to appreciating its structure. The data transformation process included converting object-type data to integers along with outlier detection and removing columns that were not useful for prediction purposes. The suitable statistical models helped us test null versus alternative hypotheses to validate

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on April 16,2026 at 04:01:33 UTC from IEEE Xplore. Restrictions apply.

our research questions through result analysis. The research team trained the chosen machine learning model after which they conducted prediction evaluations and studied the final output results. Our research project follows the workflow which is depicted in Fig 1. In addition to these processes, we applied Explainable AI Techniques (namely SHAP and LIME) to have a holistic visualization of our predictions. To interpret the model and see how different features contributed to the predictions we used both SHAP and LIME. These techniques allowed us to interpret the results of the model by observing how individual features contributed to the overall prediction.This approach helped us gain deeper insights into model behavior and ensure that our predictions were explainable.

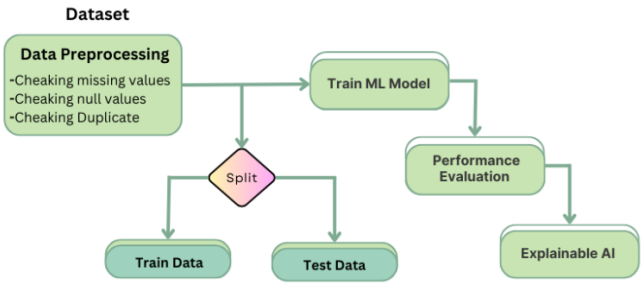

Fig. 1. Workflow Diagram

A. Data Description

For our research, we used a DataCo SMART Supply Chain dataset from DataCo Global [16], providing detailed records of supply chain activities across various product categories and customer demographics. This dataset is essential for understanding supply chain operations and evaluating factors such as shipping delays, customer satisfaction, and product performance. With 182,020 records (rows) and 53 attributes (columns), the dataset provides thorough insights into how supply chain operations operate. Among the dataset's important characteristics are:

- Days for Shipping (real): The count of days that the delivery actually took to arrive.
- Sales per Customer: The total sales made by each customer.
- Benefit per Order: The profit generated per order.
- Late Delivery Risk: A binary indicator representing the likelihood of a late delivery.
- Delivery Status: The status of the delivery, indicating whether it is "On-time," "Late," or "Advance Shipping."

The dataset reveals shipping efficiency measures and delivery delay patterns together with behavioral patterns of customers while providing numerical data about sales and delivery durations and categorizing products and customer attributes. The platform contains three main groups of products including Clothing, Sports and Electronics while uniting ordered information with untracked website activity for analytical purposes.

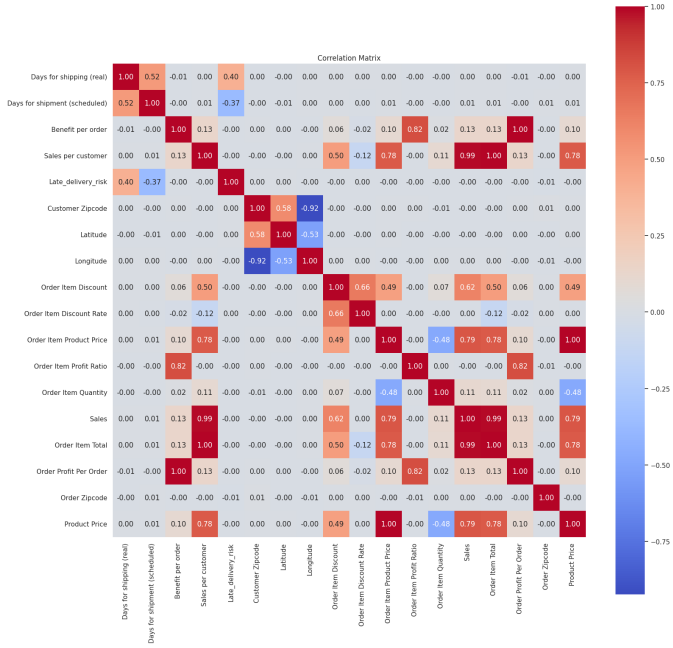

Fig. 2. Correlation Matrix

In the Exploratory Data Analysis (EDA) phase, we explored the relationship between the variables and treated missing data via continuous variable approaches (mean, median, standard deviation). Correlation, to gain insight into these relationships, we created a correlation matrix (Fig. 2) which helps you visualize how different variables correlate with each other. To evaluate these trends, test for outliers, and assess model performance, box plots and histograms were also used.

B. Statistical Tools

Several statistical tools will enable researchers to obtain important insights from supply chain dataset information which supports decision making processes. In order to derive central tendencies and distribution statistics the analysis will perform calculations of mean values and median and standard deviation and interquartile range measurements on continuous variables including lead times and costs and demand data. The distribution patterns of categorical variables will be defined using the frequency count method. A statistical technique for figuring out the link between categorical variables in a contingency table is the Chi-Square test. It evaluates if there is a substantial difference between the observed and expected frequencies of various categories under the presumption of independence. This test is commonly applied in hypothesis testing, market research, and genetics to analyze categorical data relationships. It requires assumptions such as a sufficiently large sample size and expected frequencies above a certain threshold for accuracy. While useful for categorical association, it does not measure causality and may be affected by sample size biases.

A statistical method for comparing the means of two related groups to see if there is a significant difference between them is the Paired T-Test.It is applied when the same subjects

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on April 16,2026 at 04:01:33 UTC from IEEE Xplore. Restrictions apply.

are measured before and after an intervention, ensuring that individual variability is controlled. The t-statistic, mean difference, and standard deviation are computed by the test and compared to a critical value derived from the t-distribution. It is predicated on the sample being chosen at random and the differences being regularly distributed. In experimental investigations, including medical trials, the paired t-test is frequently used to assess the effects of treatments. On the other hand, non-parametric options such as the Wilcoxon Signed-Rank Test can be better suitable if the normalcy assumption is broken.

One technique for figuring out the direction and strength of a linear relationship between two continuous variables is the Pearson correlation test. It produces a correlation coefficient, denoted by r, that falls between -1 and +1. A perfect positive linear relationship is indicated by a value of +1, which means that as one variable rises, the other one does the same consistently. Changes in one variable do not necessarily anticipate changes in the other, as indicated by a value of 0. A perfect negative linear connection, in which one variable rises while the other continuously falls, is represented by a value of -1. The variables are assumed to be continuous, to have a linear connection, to have a normal distribution, and to be free of major outliers that could distort the results. It is frequently used to study and evaluate correlations between data sets in a variety of domains, including the natural sciences, psychology, and economics.

C. Machine Learning Models

Several machine learning models were applied in this work to forecast delivery status and evaluate the likelihood of late deliveries: A basic statistical and machine learning approach for simulating the relationship between a dependent variable and one or more independent variables is called linear regression. Using the least squares approach, the objective is to reduce the discrepancy between the actual and anticipated values. Trend analysis, predictive modeling, and assessing the degree of correlation between variables are all common applications for linear regression. However, it is sensitive to outliers and assumes homoscedasticity and independence of errors.

An ensemble learning method called Random Forest Regression builds several decision trees and averages their predictions to increase precision and decrease overfitting. To ensure model variety, subsets of data and features are chosen at random to train individual trees. All decision tree outputs are averaged to produce the final prediction, which is resistant against noise and outliers. For complicated regression situations including nonlinear relationships, Random Forest works incredibly well. In addition, it offers feature importance rankings, which aid in comprehending how each feature contributes to the forecast. In contrast to simpler models, it lacks interpretability and can be computationally costly for large datasets.

Ridge Regression is a regularization method that goes beyond linear regression by preventing overfitting by including a penalty element in the cost function. To lessen the influence of less significant features, it adds an L2 regularization factor, which is the total of the squared values of the regression coefficients. It improves generalization, lowers variance, and stabilizes regression coefficients. However, it does not perform feature selection as all coefficients are shrunk but not eliminated.

A non-parametric model called Decision Tree Regression divides the dataset into subgroups according to feature values in order to generate predictions. It recursively partitions the data using conditions that minimize variance in the target variable, forming a tree-like structure. To avoid overfitting, hyperparameters such as maximum depth and minimum samples per split can be used to control the tree's depth and complexity.Decision Trees are interpretable and capture complex relationships, making them useful for nonlinear regression problems. However, they tend to overfit when grown too deep and are sensitive to small variations in data. Pruning techniques and ensemble methods like Random Forest help mitigate these issues.

D. Explainable AI (XAI)

In machine learning, SHAP is a potent explainability tool that helps interpret how specific features affect a model's predictions. Each feature is given a Shapley value based on game theory, which indicates how much it contributes to the prediction in relation to an average baseline. SHAP values help in understanding how different input variables influence the outcome, providing both local and global interpretability. It is particularly useful for complex models like Random Forest, XGBoost, and Deep Learning, where traditional feature importance measures may not suffice. SHAP is widely used in AI fairness, healthcare, and financial applications to ensure transparency and trust in model predictions. However, computing SHAP values can be computationally expensive for large datasets.

LIME is a model-agnostic interpretability technique that explains individual predictions of complex machine learning models. It works by generating synthetic perturbations around a specific data instance and training a simpler, interpretable model (like linear regression) to approximate the local decision boundary. This method provides insights into why a model made a specific prediction by identifying which features were most influential. LIME is useful for explaining black-box models such as neural networks and ensemble methods, enhancing trust and accountability in AI systems. However, its explanations are local and may not generalize well across the entire dataset. Choosing the right parameters for LIME, such as the number of perturbed samples, is crucial for reliable results.

IV. Result and Discussion

Before conducting the experiment, we have done preprocessing to ensure data quality and boost model performance. After that, the dataset was divided 80:20 for testing and training, accordingly.

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on April 16,2026 at 04:01:33 UTC from IEEE Xplore. Restrictions apply.

TABLE I
HYPOTHESIS TESTING OUTCOMES

No.	Null Hypothesis	Test Type	Statistic Value	P-Value
1	Actual vs. Scheduled Shipping Days	Paired t-Test	t-statistic = 161.236	0.0
2	Benefit per Order vs. Order Item Quantity	Pearson Correlation Analysis	Correlation coefficient (r) = 0.0157	2.57×10^{-11}
3	Sales per Customer vs. Product Price	Pearson Correlation Analysis	Correlation coefficient (r) = 0.7818	0.0
4	Late Delivery Risk vs. Shipping Mode	Chi-Square Test	Chi-square statistic = 37,716.04	0.0
5	Late Delivery Risk vs. Delivery Status	Chi-Square Test	Chi-square statistic = 180,519.0	0.0

Table 1 indicates that we used a paired t-test to determine whether the scheduled shipping days (Days for shipment (scheduled)) and actual shipping days (Days for shipping (real)) varied significantly. The p-value is 0.0 and the t-statistic is 161.236.Since the p-value is less than 0.05, the null hypothesis (H1o) is rejected. This attests to the fact that scheduled and actual shipping days differ.

We performed a Pearson correlation test to ascertain the relationship between Benefit per Order and Order Item Quantity.The Correlation coefficient (r) is 0.0157,p-value is 2.57e-11. Even though the p-value is statistically significant, the correlation coefficient is very low, we have to reject null hypothesis H2o. It means it indicates that the number of items ordered has no meaningful impact on the benefit per order.

To find out whether higher product prices correlate with higher sales per customer, we have performed a Pearson correlation test where the Correlation coefficient (r) is 0.7818 and p-value is 0.0.We can see that the correlation is strongly positive and the p-value is extremely low. So we reject the null hypothesis H3o. So we can say that higher-priced products contribute to higher sales per customer. To determine if shipping mode significantly affects the risk of late delivery, we applied Chi-Square test. We discovered that the p-value is 0.0 and the chi-square statistic is 37,716.04.We reject null hypothesis H4o since the p-value is very low.So,we can say that Shipping Mode clearly impacts Late Delivery Risk.

We performed another Chi-Square test to determine whether Late Delivery Risk is associated with Delivery Status, and the results show that the p-value is 0.0 and the Chi-square statistic is 180,519.0. We reject null hypothesis H5o because of the extremely low p-value, which also demonstrates that there is a clear correlation between late deliveries and the hazards associated with them. After testing the hypothesis, we have split the dataset into 80-20% where 80% for training and 20% for testing. From Table 2, we can see different performance metrics for different Machine Learning models in our study.

TABLE II
MACHINE LEARNING PERFORMANCE

Model Name	MSE	MAE	R^2 Score
Linear Regression	2.27×10^{-6}	0.000570	0.9999
Random Forest Regression	3.84	0.224913	0.9997
Ridge Regression	2.27×10^{-6}	0.000569	0.9999
Lasso Regression	5.07×10^{-5}	0.004320	0.9999
Decision Tree Regression	79.22	4.369237	0.9954

The regression models were assessed using MSE, MAE, and R² Score to evaluate their predictive performance. Linear Regression and Ridge Regression achieved the lowest error metrics (MSE: 2.27e-06, MAE: 0.000570 and 0.000569 respectively) and near-perfect R² Scores of 0.9999, making them highly accurate models. Lasso Regression displayed slightly higher error values (MSE: 5.07e-05, MAE: 0.004320) but maintained an impressive R² Score of 0.9999, proving effective in balancing prediction and regularization. Although less accurate than the best-performing models, Random Forest Regression produced respectably high results (MSE: 3.84, MAE: 0.224913) with a strong R2 Score of 0.9997. The highest error metrics were shown by Decision Tree Regression (MSE: 79.22, MAE: 4.369237) and the lowest R² Score of 0.9954, reflecting weaker predictive capabilities likely due to overfitting. Overall, Linear Regression and Ridge Regression emerged as the most reliable models, with Lasso Regression and Random Forest providing alternative advantages.

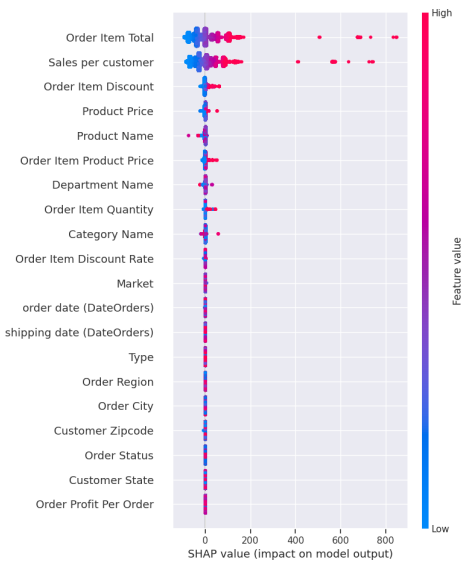

Fig. 3. SHAP Outcome

Again, we applied the SHAP and LIME using the Linear Regression model. Order Item Total, Sales per Customer, and Order Item Discount are the top features that significantly

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on April 16,2026 at 04:01:33 UTC from IEEE Xplore. Restrictions apply.

impact the model's output, as shown in Fig. 3. Red denotes greater values and blue denotes lower values in the color gradient that shows feature values. We have used SHAP values to analyze feature importance in our model.

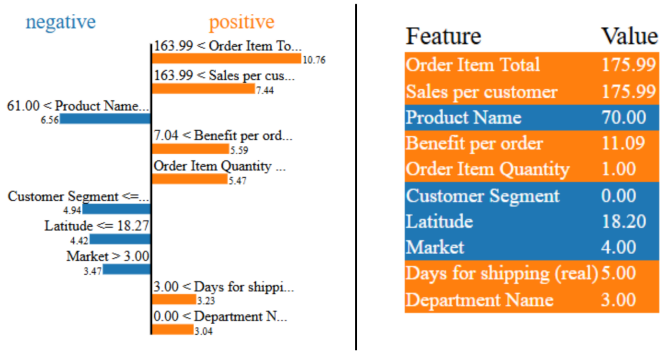

Fig. 4. Lime Outcome

On the other hand, we have used LIME because it provides explanations for individual predictions rather than overall feature importance. From Fig. 4, we have found out that the top positive contributors to the specific prediction are Order Item Total, Sales per Customer, and Product Name. On the other hand, features like Customer Segment and Days for Shipping (real) slightly abbreviate the predicted value.

V. Conclusion

This work demonstrates that coupling rigorous hypothesis testing with explainable machine-learning models can forecast delivery delays with near-perfect accuracy and still provide managers with clear operational insight. Among the six learners evaluated, Ridge Regression achieved the lowest error metrics ($MSE = 2.27 \times 10^{-6}$, $MAE = 5.7 \times 10^{-4}$, $R^2 = 0.9999$), demonstrating that a regularised linear model can rival more complex ensembles when feature engineering is sound. SHAP and LIME explanations revealed that *Order-Item-Total*, *Sales-per-Customer*, and *Order-Item-Discount* are the main drivers of late-delivery risk, thereby closing the interpretability gap that often hampers adoption of data-driven tools in logistics.

However,the study has limitations; it uses only one industry-specific dataset, and it does not consider any outside influences, such as the weather, time of year, labor strikes or port congestion, all factors that frequently lead to real-world delays. Additionally, analysis was performed on historical batch data, as opposed to real-time, continuous data as in the current work, and the financial impact of mitigation measures due to model predictions was not quantified.

To bridge these gaps, we propose future studies that incorporate online learning architectures that adapt to concept drift, integrate multimodal predictors such as traffic feeds and IoT sensors, and embed the delay-prediction module within prescriptive dashboards. These dashboards would not only forecast delays but also recommend proactive countermeasures and justify their cost-effectiveness.By addressing these extensions, the model presented in this study can be extended to a full-fledged decision support system which may provide logistics managers the feasible solutions that are required in assessing the risk, cost and in correlating it with customer satisfaction within the existing supply chain systems.

References

[1] A. Rejeb, J.G. Keogh, and S.F. et al. Wamba. The potentials of augmented reality in supply chain management: a state-of-the-art review. *Manag Rev Q*, 71:819–856, 2021.

[2] A. Belhadi, S. Kamble, S. Fosso Wamba, and M. M. Queiroz. Building supply-chain resilience: an artificial intelligence-based technique and decision-making framework. *International Journal of Production Research*, 60(14):4487–4507, 2021.

[3] Yang Lei, Hou Qiaoming, and Zhao Tong. Research on supply chain financial risk prevention based on machine learning. *Computational Intelligence and Neuroscience*, pages 6531154, 15 pages, 2023.

[4] A. Aljohani. Predictive analytics and machine learning for real-time supply chain risk mitigation and agility. *Sustainability*, 15(20):15088, 2023.

[5] X. Zhu, A. Ninh, H. Zhao, and Z. Liu. Demand forecasting with supply-chain information and machine learning: Evidence in the pharmaceutical industry. *Production and Operations Management*, 30(9):3231–3252, 2021.

[6] S. Khan, A.Z. Piprani, and Z. Yu. Digital technology and circular economy practices: future of supply chains. *Oper Manag Res*, 15:676–688, 2022.

[7] S.F. Nimmy, O.K. Hussain, R.K. Chakrabortty, F.K. Hussain, and M. Saberi. Explainability in supply chain operational risk management: A systematic literature review. *Knowledge-Based Systems*, 235:107587, 2022.

[8] S. Tiwari, P. Sharma, T. M. Choi, and A. Lim. Blockchain and third-party logistics for global supply chain operations: Stakeholders' perspectives and decision roadmap. *Transportation Research Part E: Logistics and Transportation Review*, 170:Article 103012, 2023.

[9] P. Helo and Y. Hao. Artificial intelligence in operations management and supply chain management: an exploratory case study. *Production Planning & Control*, 33(16):1573–1590, 2021.

[10] Mohd Maaz and Rais Ahmad. Impact of supply chain performance on organizational performance mediated by customer satisfaction: A study of dairy industry. *Business Process Management Journal*, 2022.

[11] Muhammad Hunaid, M Bhurgri, and A. Shaikh. Supply chain visibility in leading organizations of the shipping industry. *South Asian Journal of Social Review*, 1:8–20, 2022.

[12] Md Rauf, Md Majadul Islam Jim, Md Rahman, and Md. Tariquzzaman. Ai-powered predictive analytics for intellectual property risk management in supply chain operations: A big data approach. *Global Mainstream Journal*, 1:32–46, 2024.

[13] H. Zhou, Q. Wang, and Q. Yang. How does digitalisation influence supply chain performance? evidence from a supply chain risk management perspective. *International Journal of Logistics Research and Applications*, 28(3):325–343, 2023.

[14] Wang Fang, Hengjie Zhang, and Jigan Wang. Strategic behavior in multi-criteria sorting with trust relationships-based consensus mechanism: Application in supply chain risk management. *European Journal of Operational Research*, 321, 2024.

[15] J. Tao, M. Aamir, M. Shoaib, N. Yasir, and M. Babar. Bridging the gap between supply chain risk and organizational performance conditioning to demand uncertainty. *Sustainability*, 17(6):2462, 2025.

[16] Alino Rani Anesfahani. Dataco smart supply chain for big data analysis. https://www.kaggle.com/datasets/alinoranianesfahani/dataco-smart-supply-chain-for-big-data-analysis, 2019. Accessed: 2025-04-26.

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on April 16,2026 at 04:01:33 UTC from IEEE Xplore. Restrictions apply.